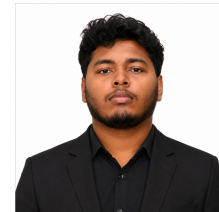


DEBABRATA SAHU

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Berhampur, Odisha, India



PROFILE

Electronics and Communication Engineering undergraduate with research and project experience in digital VLSI, RTL design, FPGA-based systems, and electronic circuit design. Interested in research in VLSI and IC design, hardware-efficient computing, RTL design and verification, FPGA architectures, and semiconductor technologies. Experienced in Verilog, Xilinx Vivado, Cadence Virtuoso, and CST Studio Suite through academic projects, research, and technical training.

EDUCATION

NIST University B.Tech – Electronics and Communication Engineering — CGPA: 7.7/10	<i>2023 – 2027</i> Berhampur, India
Government Autonomous College Intermediate – Science — Percentage: 70%	<i>2020 – 2022</i> Rourkela, India

PROJECTS

Approximate Multiplier for AI Applications	<i>08/2026 – Present</i>
Designed and implemented an approximate multiplier in Verilog for hardware-efficient arithmetic in AI-oriented applications. Synthesized and evaluated the RTL design using Xilinx Vivado with emphasis on hardware resources, timing, power, and approximation error.	
Configurable Floating-Point FMA Unit using Mixed Precision	<i>02/2026 – 04/2026</i>
Designed a configurable Floating-Point Fused Multiply-Add (FMA) unit using mixed precision for hardware-efficient floating-point computation. Developed and simulated the RTL implementation using Verilog in Xilinx Vivado, focusing on configurable precision and arithmetic functionality.	
Terahertz Metamaterial Absorber for Skin Cancer Detection	<i>05/2026 – 06/2026</i>
Designed and simulated a terahertz metamaterial absorber-based sensing system using CST Studio Suite for potential skin cancer detection. Analyzed electromagnetic response characteristics of the absorber to study changes associated with abnormal tissue and evaluate sensing behavior.	
12-Bit Analog-to-Digital Converter	<i>08/2025 – 09/2025</i>
Designed and simulated a 12-bit Analog-to-Digital Converter (ADC) using Cadence Virtuoso for analog-to-digital signal conversion. Performed schematic-level design and simulation to analyze converter functionality and performance.	
Smart Greenhouse Monitor System	<i>05/2025 – 06/2025</i>
Developed an IoT-based greenhouse monitoring system integrating temperature, humidity, gas, and motion sensors for real-time environmental monitoring. Implemented automated control using servo motors, buzzer, LED, Wi-Fi, and Blynk for remote monitoring and plant-safety alerts.	
Smart Mining Helmet	<i>03/2025 – 04/2025</i>
Developed a sensor-integrated mining helmet for monitoring worker heart rate, temperature, humidity, and hazardous gas conditions. Implemented Wi-Fi-based cloud monitoring using ThingSpeak and Blynk with automated alerts through buzzer, LED, and LCD.	

INTERNSHIP & RESEARCH EXPERIENCE

Research Intern National Institute of Technology Jamshedpur	<i>18 May 2026 – 18 June 2026</i> Jamshedpur, Jharkhand
Gained research exposure in terahertz technology, metamaterial-based sensing, and electromagnetic simulation for biomedical applications.	
VLSI Project-Based Intern EdXcellence	<i>20 January 2026 – 18 March 2026</i>
Gained hands-on exposure to VLSI design, RTL development, digital logic, and hardware-oriented design methodologies using Verilog/SystemVerilog and Xilinx Vivado.	

E&T Department Intern

Central Coalfields Limited

01 June 2025 – 30 June 2025

Ranchi, Jharkhand

Gained practical exposure to embedded systems, sensor interfacing, IoT communication, and real-time monitoring through industrial technology applications.

SKILLS

HDL & Programming: Verilog, Python, C, MATLAB, 8086 Assembly

EDA & Simulation Tools: Xilinx Vivado, Cadence Virtuoso, CST Studio Suite, MATLAB, EDA Playground

Digital & Hardware: Digital VLSI Design, RTL Design, RTL Verification, FPGA, Digital Logic Design, Hardware Debugging

Embedded & IoT: Arduino, ESP32, Arduino IDE, Sensor Interfacing, ThingSpeak, Blynk, Microcontrollers, 8086 Microprocessor

TRAINING & CERTIFICATIONS

FPGA-Based Communication System Design (FCSD-2026)

25 May 2026 – 05 June 2026

Hands-on training in FPGA-based system design, digital communication systems, hardware implementation, and FPGA architecture and development.

Getting Started with Artificial Intelligence

27 February 2026

Completed introductory training in artificial intelligence concepts and applications through IBM SkillsBuild.

MATLAB & Its Applications in Engineering

12 July 2024 – 01 August 2024

Gained practical exposure to MATLAB for engineering computation, analysis, simulation, and problem-solving.

WORKSHOPS & TECHNICAL LEADERSHIP

Tech Spark – IoT Workshop

18–19 September 2026

Organized a two-day hands-on IoT workshop through the Renewable Energy Club, focusing on practical applications of IoT, sensors, and embedded technologies.

SensoTech – Hands-on Electronics Workshop

11–12 April 2025

Organized a two-day hands-on workshop through the Renewable Energy Club covering sensors, microcontrollers, project building, and practical applications of electronics.

Tinker to Tech – Project Building Workshop

26–27 August 2024

Participated in a two-day hands-on workshop organized by the Renewable Energy Club in collaboration with the IEEE Student Branch, NIST University, covering sensors, microcontrollers, and basic electronics project development.

EXTRACURRICULAR ACTIVITIES

Renewable Energy Club

September 2024 – Present

Event Head • Social Media Head

NIST University, Berhampur

Innovation & Incubation Club

November 2025 – Present

Core Member

NIST University, Berhampur

NIST Basketball Club

October 2023 – Present

Player

NIST University, Berhampur

REFERENCES

Dr. Sandipan Mallik

Professor, Head of Department of Electronics and Communication Engineering

NIST University

✉ sandipan@nist.edu ☎ +91-9658965687

Dr. Ashwini Kumar Nayak

Sr. Assistant Professor, Department of Electrical Engineering

NIST University

✉ ashwini.nayak@nist.edu ☎ 91-7978587547

DECLARATION

I hereby declare that the information given above is true and accurate to the best of my knowledge and belief.

Berhampur, Odisha

Debabrata Sahu